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1 字串

1.1 字典樹 trie

```

1 struct trie{
2     trie *nxt[26]={};
3     int cnt=0;
4     int sz=0;
5 };
6 trie *root=new trie();
7 void insert(const string &s){
8     trie *now=root;
9     for(auto i:s){
10         if(now->nxt[i-'a']==NULL){
11             now->nxt[i-'a']=new trie();
12         }
13         now=now->nxt[i-'a'];
14     }
15 int qry(string &s){
16     trie *now=root;
17     for(auto i:s){
18         if(now->nxt[i-'a']==NULL) return;
19         now=now->nxt[i-'a'];
20     }
21     return now->cnt;
22 }

```

1.2 KMP

```

1 vector<int> kmp(string s,string t){
2     if(t.size()>s.size()) return {};
3     vector<int> pi(t.size(),0);
4     for(int i=1,j=0;i<t.size();i++){
5         while(j>0 && t[j]!=t[i]){
6             j=pi[j-1];
7         }
8         if(t[j]==t[i]) j++;
9         pi[i]=j;
10    }
11    vector<int> ans;
12    for(int i=0,j=0;i<s.size();i++){
13        while(j>0 && t[j]!=s[i]){
14            j=pi[j-1];
15        }
16        if(t[j]==s[i]) j++;
17
18        if(j==t.size()){
19            ans.push_back(i-j+1);
20            j=pi[j-1];
21        }
22    }
23    return ans;

```

2 最短路徑

2.1 basic

```

1 // c++ code
2 #include <bits/stdc++.h>

```

```

3 using namespace std;
4

```

```

1 int main() {
2     // test comment
3     cout << "test string\n";
4 }

```

3 圖論

3.1 凸包問題

```

1 struct node{
2     int x,y;
3     bool operator<(const node &other) const{
4         return
5             (x<other.x) || (x==other.x && y<other.y);
6     };
7 };
8 struct ConvexHull{
9     vector<node> vec;
10    int n;
11    void init(vector<node> &v){
12        vec=v;
13        n=v.size();
14    }
15    int cross(node a,node o,node b){
16        return
17            (a.x-o.x)*(b.y-o.y)-(b.x-o.x)*(a.y-o.y);
18    }
19    vector<int> point;
20    vector<int> area;
21    void build(){
22        point=vector<int>(n+1);
23        area=vector<int>(n+1);
24        vector<node> up,down;
25        int upsum=0;
26        int downsum=0;
27        for(int i=0;i<n;i++){
28            while(up.size()>=2 &&
29                cross(up[up.size()-2],
30                    up[up.size()-1],
31                    vec[i])<=0){
32                upsum-=cross(up[up.size()-2],
33                            {0,0},up[up.size()-1]);
34                up.pop_back();
35            }
36            up.push_back(vec[i]);
37            if(up.size()>1)
38                upsum+=cross(up[up.size()-2],
39                            {0,0},
40                            up[up.size()-1]);
41
42            while(down.size()>=2 &&
43                cross(down[down.size()-2],
44                    down[down.size()-1],
45                    vec[i])>=0){
46                downsum-=cross(down[down.size()-2],
47                            {0,0},down[down.size()-1]);
48                down.pop_back();
49            }
50            down.push_back(vec[i]);
51            if(down.size()>1)
52                downsum+=cross(down[down.size()-2],
53                            {0,0},down[down.size()-1]);
54
55            area[i+1]=abs(upsum-downsum);
56        }
57    }

```

```
58         if(i+1==1) point[i+1]=1;
59         else point[i+1]=up.size()+down.size()-2;
60     }}
61     double getarea(int p){
62         return area[p]/2.0;
63     }
64     int getnum(int p){
65         return point[p];
66     }
67 };
```

4 數學

4.1 公式

- 中文測試

- $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$